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Experiment-4

Capture and Analyse network traffic using wireshark

AIM:

To capture, save and analyse network traffic including TCP, SDP, IP, HTTP, ARP, DHCP, ICMP & DNS protocols using wireshark network analyzer tool.

REQUIREMENT:

Wireshark is installed, linux or windows Administrative privilege, active network connection web browser.

PROCEDURE:

1) Install wireshark: sudo apt install wireshark
2) Launch wireshark with admin privileges.
3) select the active network interface
4) start capture by clicking the shark-fin button
5) Generate traffic: Open a browser, ping a host, run nslookup.
6) Stop capture after 30 secs.
7) Filter TCP traffic: type 'tcp' in the display filter bar.

Commands

install wireshark on linux

sudo apt-get install wireshark -y

sudo usermode -aG wireshark $USER

capture with Tshark

sudo tshark -i eth0 -c 100 -w capture.pcap

display filters

tcp # All tcp traffic

udp # All UDP traffic

http # HTTP only

dns # dns queries/responses

arp # ARP traffic

icmp # ping traffic

dhcp # DHCP DORA proces

ip.addr == 8.8.8.8 # traffic

tcp.port == 443 # HTTPS traffic

http.request.method == GET # HTTP get request

tshark analysis

tshark -r capture.pcapng -Y 'http' -T fields -e ip.src

tshark -r capture.pcapng -Y 'dns.qry.name' -T fields

tshark -r capture.pcapng -q -z io,phs # protocol hierarchy

8) Find HTTP traffic : filter 'http' and follow tcp stream

9) Filter DNS : type DNS and examine query response

10) Filter ARP : type 'arp' and view Mac-IP resolution

11) Filter ICMP : type ICMP and observe ping packets.

12) Filter DHCP : type 'dhcp' and examine DORA process.

13) Export the capture file as .pcapng

14) use statistics → protocols hierarchy to see traffic distribution.

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RESULT:

Network traffic was successfully captured and analyzed using wireshark. TCP handshake, DNS queries, HTTP requests, ARP resolution, DHCP lease process and ICMP exchanges were identified and examined.